WBSL BRIDGE

Intent-Aware Real-Time Bidirectional Sign Language Communication System for the Deaf Community of West Bengal

Multimodal Deep Learning · Constrained LLMs · Hybrid Cloud–Edge Inference

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The Problem: A Silent Technological Void

70M+

Deaf & hard-of-hearing individuals rely on sign language worldwide
Across India, the 2011 Census recorded roughly 50 lakh people with hearing disabilities

170

Signs only — the entire static WBSL lexical resource (Wikisigns)

Near 0~

Dedicated AI systems exist for West Bengal Sign Language today

WBSL is linguistically distinct from Delhi ISL and Bangladesh BdSL (Johnson & Johnson, 2016) — yet remains entirely unserved by modern AI, while Google DeepMind's SL2T (Aug 2026) proves consumer-scale ASL translation is possible.

The Intention Problem — most systems translate word-by-word, losing the signer's true intent. A raised eyebrow signals a question; a head shake signals negation. Ignoring these non-manual markers can invert the meaning entirely.

Literature Survey

Progression from sensor-based recognition to national-scale continuous translation

Work	Contribution	Limitation
Sarker & Hoque (2018)	Glove-based BdSL conversion	*Hardware-dependent; blind to facial expressions*
Islam & Mousumi — Ishara-Lipi	First open-access isolated Bangla sign dataset	*Static characters only; no continuous signing*
Akash et al. (2023)	Real-time BdSL sentence formation via action recognition	*Bangladesh BdSL only*
SignISL / Signer.AI	National ISL tools (Indian Railways, IIIT Bangalore)	*Standard ISL → Hindi/English only; no WBSL*
ISLTranslate / iSign (ACL 2024)	Continuous pan-Indian ISL translation datasets	*Output limited to English*
Google DeepMind SL2T (Aug 2026)	Real-time ASL-to-English at consumer scale, 100,000+ hrs data	*Targets ASL; regional low-resource languages left behind*

Identified Research Gaps

G1
Non-Manual Marker Gap
Facial/head grammar cues under-modeled

G2
Regional Linguistic Gap
AI targets standard ISL/BdSL, not WBSL

G3
Data Gap
Only 170 signs; no continuous WBSL dataset

G4
Grammar & Structure Gap
Sign glosses need SOV-aware translation

G5
Resource-Constrained Deployment Gap
Heavy models unusable on CPU-only devices

G6
Real-Time Latency Gap
Low-latency pipeline remains an engineering challenge

G7
Bidirectional Gap
No dedicated two-way WBSL communication system

Proposed Model — Forward Path (Sign → Bengali)

Webcam
30 FPS

→

MediaPipe
Holistic
540 landmarks

→

DeepFace +
Geometry NMM +
Temporal LSTM

→

LLM Inference
(Cloud/Local)

→

Bengali Text
+ Piper/gTTS

Three Parallel Modules Feed Intent into the LLM

Emotion (DeepFace)

Happy, Sad, Angry, Neutral — 7-class classification

Grammar NMM (Geometry)

Eyebrow raise → Question · Head shake → Negation

Sign Gloss (Temporal LSTM, ONNX)

Continuous hand/body movement → gloss sequence, <5ms/seq

Bengali Text Generation

Handles SOV word order · Agglutination · Honorifics — gloss + intent reconstructed into grammatically correct Bengali

Proposed Model — Reverse Path (Bengali → Sign)

Key Architectural Innovations

1 Multimodal Intent Capture

Hands + face + body + emotion + grammar markers all feed the LLM

2 User-Selectable LLM Architecture

Cloud OpenRouter or local KoboldCpp, chosen per speed/privacy/internet

3 Privacy by Design

Raw video discarded — only landmarks flow through the pipeline

4 ONNX-Accelerated Inference

LSTM runs at under 5ms per sequence on CPU

LLM Inference Strategy — Measured Benchmarks

User-selectable architecture — no automatic switching; the user controls the trade-off

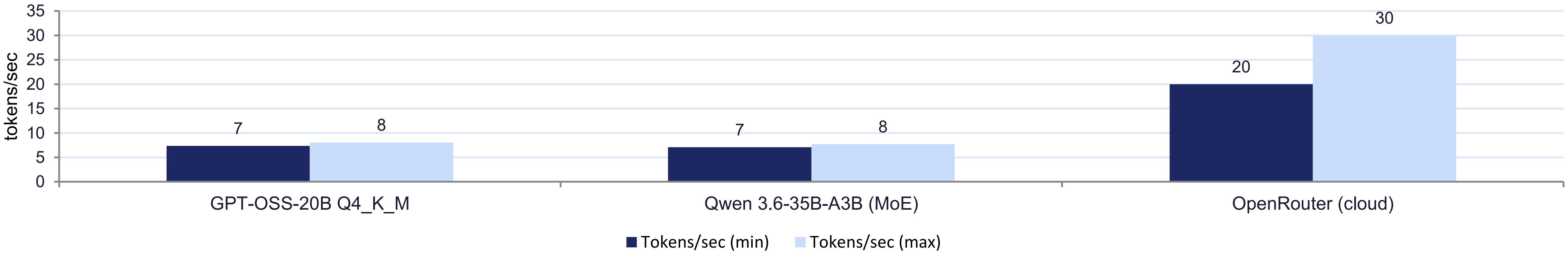
Cloud — OpenRouter API

- Higher generation speed
- Access to powerful cloud models
- Requires internet connectivity
- Speed: ~20–30 tokens/sec

Local — KoboldCpp

- Fully offline after setup
- Greater privacy — data stays on device
- Quantized GGUF models
- Speed: ~7–20 tokens/sec (Ryzen 7U, 16GB)

Benchmark Chart — Local Model Throughput (tokens/sec, Ryzen 7 U, 16GB RAM)



Methodology & Technology Stack

Dataset Strategy

- Primary training: ISLTranslate & iSign (ISL taught in WB schools)
- Baselines: Ishara-Lipi, BdSLW60 for isolated recognition (but not preferring BdSL openly)
- Community collection: 50–100 WBSL sentences with Deaf participants (consent-based)
- Supplementary WBSL video review from public platforms for pattern study & annotation, where licensing allows

Minimum Initial Data

•**WBSL:** 100–200 signs

•**WBSL sentences:** 100–300

•**ISL:** 1,000+ samples/sentences

Technology Stack

Vision: MediaPipe Holistic + OpenCV	**Emotion:** DeepFace
Temporal Model: PyTorch LSTM + ONNX Runtime	**LLM (Cloud):** OpenRouter API
LLM (Local): KoboldCpp + GGUF	**Reverse Backend:** FastAPI + NLP/LLM
Reverse Frontend: React UI	**TTS:** Piper (offline) / gTTS (online)

Training Experiments:

```
Model A → WBSL only

Model B → ISL → WBSL Fine-tuning

Model C → ISL + WBSL
```

Main target: WBSL
Supporting data: ISL
Important: Keep ISL and WBSL separately labelled.

Project Timeline

Proposed workflow across the remaining project phase

M1	**Data Collection & Setup**	WBSL community sentences, MediaPipe pipeline, dataset consolidation
M2	**Model Development**	Temporal LSTM training, DeepFace + NMM geometry integration
M3	**LLM Integration**	OpenRouter + KoboldCpp pipelines, Bengali grammar reconstruction
M4	**Reverse Path Build**	React UI + FastAPI backend, sign gloss mapping & animation
M5	**Evaluation & Testing**	Signer independence, BLEU/BERTScore, Deaf community trials
M6	**Refinement & Deployment**	Latency optimization, final benchmarking, documentation

Evaluation Framework & Expected Results

Evaluation Metrics

- Recognition: Accuracy, Precision, Recall, F1 on held-out signers
- Translation: BLEU, BERTScore, chrF++ for Bengali quality
- Intent Preservation: Deaf signers + hearing raters on 50 test sentences
- Signer Independence: trained/tested on disjoint signers, target ≥80% accuracy

Expected Performance

≥30 FPS
MediaPipe tracking

<50 ms
Emotion detection latency

<5 ms
LSTM inference / sequence

1.5–2.5 s
End-to-end (cloud) / sign

3–4 s
End-to-end (local) / sign

<12 GB
Total RAM usage

Future Work

1 Expand WBSL Dataset

Grow community-recorded corpus beyond initial 50–100 sentences for stronger generalization

2 Avatar-Based Sign Generation

Move from clip-mapping to a parametric 3D avatar for smoother reverse-path animation

3 Mobile & Edge Deployment

Optimize the pipeline for smartphones and low-power edge devices

4 Regional Dialect Coverage

Extend beyond WBSL to neighbouring low-resource regional sign variants

5 Field Trials with Deaf Community

Longitudinal usability studies across West Bengal to refine intent accuracy

6 Adaptive LLM Switching

Explore context-aware auto-selection between cloud and local inference

Conclusion

"Can we build a system that understands the intention of a linguistically distinct, data-scarce Deaf community — and speaks back to them in their own language, on their own hardware?"

- Combines MediaPipe, DeepFace, geometric NMM analysis, temporal LSTM, and user-selectable cloud/local LLM inference
- Translates meaning, not just words — preserving question, negation, and emotional intent
- Works offline and online, preserving privacy via landmark-only processing
- Aims to give West Bengal's Deaf community a voice in their own language

Thank You

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